

Chapter 7 Continuity

7.1 Limits and continuity

A function f is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$.

Equivalently: for every $\epsilon > 0$ there is a $\delta > 0$ such that

$|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$.

The order of the quantifiers matters: δ is chosen after ϵ and after a .

7.1 Limits and continuity

Example. $f(x) = x^2$ is continuous at every a .

Given $\epsilon > 0$, choose $\delta = \min(1, \epsilon / (2|a| + 1))$.

Then $|x^2 - a^2| = |x - a| |x + a| < \delta (2|a| + 1) \leq \epsilon$.

Notice that δ depends on a as well as on ϵ .

7.2 Three hard theorems

Theorem 1. If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$, then there is some x in $[a, b]$ with $f(x) = 0$.

Theorem 2. If f is continuous on $[a, b]$, then f is bounded above on $[a, b]$.

Theorem 3. If f is continuous on $[a, b]$, then there is some y in $[a, b]$ such that $f(y) \geq f(x)$ for all x in $[a, b]$.

7.2 Three hard theorems

The intermediate value theorem follows from Theorem 1 applied to $f - c$.

These theorems fail on open intervals: $f(x) = 1/x$ on $(0, 1)$ is unbounded.

The proofs require the least upper bound property of the real numbers.

7.3 Uniform continuity

Definition. f is uniformly continuous on an interval A if for every $\epsilon > 0$ there is some $\delta > 0$ such that, for all x and y in A , if $|x - y| < \delta$, then $|f(x) - f(y)| < \epsilon$.

Here δ may depend on ϵ but not on x or y .

7.3 Uniform continuity

Theorem. If f is continuous on $[a, b]$, then f is uniformly continuous on $[a, b]$.

The function $f(x) = 1/x$ is continuous but not uniformly continuous on $(0, 1)$.

The closed interval is essential: compactness lets a single delta work everywhere.

Chapter 8 Least Upper Bounds

8.1 The least upper bound property

Every nonempty set of real numbers that is bounded above has a least upper bound.

This property distinguishes the real numbers from the rational numbers.

The set $\{ x : x^2 < 2 \}$ has no least upper bound in the rationals.

8.1 The least upper bound property

Using the least upper bound property we can now prove Theorem 1 of Chapter 7.

Let $A = \{ x \text{ in } [a, b] : f \text{ is negative on } [a, x] \}$ and let $\alpha = \sup A$.

Problems. 1. Prove that α is in (a, b) . 2. Prove that $f(\alpha) = 0$.